

ECON 0100 | Exercise B6 | International Trade

I Solutions

Answers and work are shown in the red blocks.

Pumpkin pasties again, with the same demand and supply curves as Exercise B3:

$$P = 12 - \frac{Q_d}{2} \qquad P = 2 + \frac{Q_s}{2}$$

Prices are in galleons and quantity is in pasties. The equilibrium you found is 10 pasties at 7 galleons.

Q1 | Unlocking the Magic

For nearly all of recent history the Ministry of Magic required {++pumpkin pasties++} ~~broomsticks~~ to be made and purchased locally. The ministry got wind of some economics ideas around opening the domestic market to international trade and has hired you to offer {++advice++} ~~advise~~ on the impact this policy would have on the domestic {++pasty++} ~~broomstick~~ market. The world price for {++pumpkin pasties++} ~~enchanted broomsticks~~ is 5 galleons.

Use a market model to analyze the impact. Give special focus to the welfare impacts on buyers and sellers. {++Do calculate the welfare measures,++} ~~No need to calculate welfare measures. Do calculate~~ {++and++} the impact to prices and quantities.

- a) Imports or exports? **imports**, exports
- b) Quantity demanded: **14** Quantity supplied: **6** Imports: **8**
- c) Consumer surplus before: **25** after: **49**
- d) Producer surplus before: **25** after: **9**
- e) Change in total surplus: **+8**
- f) Buyers: **gain**, lose Sellers: gain, **lose**

Closed market: 10 pasties at 7 galleons, $CS = PS = 25$.

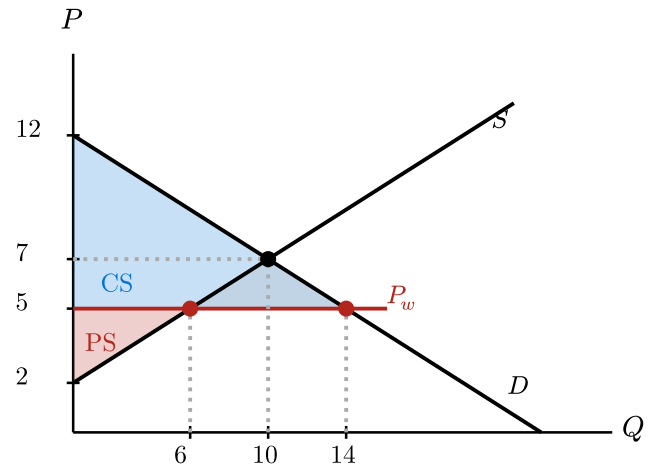
At the world price of 5, below the domestic price, buyers buy abroad: the market **imports**.

Quantity demanded 14, quantity supplied 6; imports $14 - 6 = 8$.

$CS = \frac{1}{2} \cdot 14 \cdot (12 - 5) = 49$ (+24). $PS = \frac{1}{2} \cdot 6 \cdot (5 - 2) = 9$ (-16).

Total surplus rises by 8, the triangle $\frac{1}{2} \cdot 8 \cdot |7 - 5|$ between the curves and the world price.

Buyers gain more than sellers lose.



Q2 | An Alternative World

If instead of a price of 5 galleons in Question 1 the global price had been 9 what would have been the distributional impact of opening the domestic market for trade? No need to do math here. Simply use a graph to show the impact on the market.

- a) Imports or exports? imports, **exports**
- b) Buyers: gain, **lose** Sellers: **gain**, lose
- c) Total surplus: **rises**, stays the same, falls

At 9, above the domestic price of 7, sellers sell abroad and buyers face the higher price. Domestic buyers buy 6, sellers sell 14, and 8 are exported.

Buyers lose (CS falls from 25 to 9); sellers gain (PS rises from 25 to 49); total surplus rises by 8. The mirror image of Q1: whoever is on the side that trades abroad gains.

